

Naman AGENTIC SYSTEM ARCHITECT

PROFESSIONAL SUMMARY Highly skilled and results-driven **Agentic System Architect** with 2+ years of intensive hands-on experience driving the design, architecture, and deployment of **autonomous AI agents and multi-agent systems**. Expert in **LLM orchestration** using frameworks like **LangGraph, LangChain, and LlamaIndex** (with exposure to AutoGen/CrewAI concepts), implementing complex workflows, and integrating **Large Language Models** (GPT-4, Claude, Gemini Pro, Llama 3.1) with external tools, databases, and APIs for multi-step task execution. Proficient in **Python (Expert)** for scalable AI solutions, with a proven track record in **LLMOps** (ensuring reliability, low latency, and cost-efficiency), building systems with **short-term and long-term memory**, and establishing robust **observability frameworks** to monitor agent behavior, diagnose anomalies, and refine policies. Experienced with **vector databases (ChromaDB, Pinecone)** and **cloud platforms (AWS, GCP)**, ensuring production readiness and continuous improvement.

CONTACT

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SKILLS

- **Agentic AI & LLM Frameworks:** LangGraph, LangChain, LlamaIndex, LangFlow, AutoGen (Exposure), CrewAI (Exposure)
- **LLMs & Generative AI:** Google Gemini Pro, OpenAI (GPT-4), Llama 3.1, LLM Finetuning, Transformers, Generative AI
- **Agentic System Design:** Multi-Agent Systems, Autonomous Agents, Tool Use, Function Calling, API Orchestration, State Management, Short-term & Long-term Memory, Observability Frameworks, Anomaly Diagnosis, Policy Refinement
- **LLMOps:** Deployment, Reliability, Low Latency, Cost-Efficiency, Monitoring, Production Readiness
- **Programming Languages:** Python (Expert)
- **Vector Databases:** Pinecone (Experience), ChromaDB (Experience)
- **Cloud Platforms & Containerization:** AWS (Deployment Strategies), GCP (Deployment Strategies), Azure (Familiarity), Docker
- **Other Tools & Libraries:** PyPDF2/Marker, Regex, Gradio, OpenAI Whisper, PyAudio, Text-to-Speech (TTS), Flask, OpenAPI Specification, SQL, Machine Learning Fundamentals

APPLIED EXPERIENCE

- **AI/ML Engineer (Internship) at Internselite:** July 2024 – Present
 - Contributed to foundational data engineering and LLM optimization, directly supporting the development of robust AI systems and ensuring high-quality data readiness for agentic deployments.
 - Engineered data pipelines to process and analyze over 50 GB of unstructured data, ensuring high-quality inputs for Large Language Model (LLM) fine-tuning and optimizing data readiness for agentic decision-making.
 - Partnered with cross-functional teams to iterate model training strategies,

achieving a **15% improvement** in LLM accuracy on key benchmarks, directly enhancing model performance for reliable agentic systems.

- Developed automated scripts to clean and validate training datasets, reducing data noise by **20%** and improving the consistency of model outputs, a critical step for robust agentic decision-making and policy refinement.

KEY PROJECTS & PORTFOLIO

• AI-DRIVEN RESUME OPTIMIZER (AGENTIC WORKFLOW)

- **Dates:** December 2023 – January 2024
- **Technical Stack:** Python, Google Gemini Pro, LangChain, Regex, PyPDF2/Marker, ChromaDB (Vector Store)
- Engineered an **autonomous agentic workflow** using Google Gemini Pro to iteratively optimize resumes against Job Descriptions (JD), achieving a target ATS score of **95%+** and improving average scores from 65 to 90+ across 20+ test cases.
- Developed a multi-stage feedback loop including PDF-to-Markdown extraction, scoring, and an "Autocorrection" module that refines content based on specific JD keywords and semantic gaps, demonstrating **short-term memory** and **continuous improvement**.
- Implemented an iterative refinement logic running up to 5 autonomous cycles, using LLM-based reasoning to evaluate resume-JD alignment and rewrite bullet points for maximum impact, showcasing **agent behavior monitoring** and **policy refinement**.
- Integrated a **vector database (ChromaDB)** to store and retrieve JD embeddings for semantic matching, enhancing resume optimization relevance and serving as **long-term memory**.
- Developed a final rendering pipeline to convert optimized Markdown back into ATS-friendly PDF format, ensuring both content and structural compliance for **production readiness** and **reliability**.

• MULTI-AGENT COLLABORATIVE SYSTEM

- **Dates:** January 2024 – February 2024
- **Technical Stack:** Python, LangGraph, Google Gemini, OpenAI (GPT-4), React (JSX), Flask, PyWebView
- Architected a stateful **multi-agent system** using **LangGraph** to automate the software development lifecycle, coordinating five specialized **AI agents** (Requirement Analyst, Planner, UI/UX Designer, etc.), demonstrating **complex workflow implementation**.
- Implemented a Directed Acyclic Graph (DAG) to manage complex information flow, enabling autonomous requirement engineering, gap analysis, and strategic roadmap generation, demonstrating **complex workflow implementation**.
- Engineered an "Intelligent Gap Analysis" node that identifies inconsistencies in user prompts and triggers a Human-in-the-Loop feedback cycle for **100% requirement accuracy**, serving as an **observability framework** to **diagnose anomalies** and **refine policies**.
- Developed an automated UI/UX pipeline translating high-level technical specifications into functional, responsive React (JSX) code with a live preview interface using Flask, showcasing **integration with external tools and APIs**.
- Optimized model performance by integrating a multi-LLM strategy (Gemini Flash for rapid analysis, GPT-4 for complex reasoning) with considerations for **cost-efficiency** and future **cloud deployment (AWS/GCP)**.

- **LOCAL-FIRST AGENTIC VOICE ASSISTANT**

- **Dates:** February 2024 – Present
- **Technical Stack:** Python, Llama 3.1 (Ollama/Local), OpenAI Whisper, LangGraph, PyAudio, Text-to-Speech (TTS), ChromaDB (Vector Store)
- Developed a privacy-centric, **low-latency** voice assistant capable of executing complex tasks locally, utilizing OpenAI Whisper for high-accuracy speech-to-text (STT) via PyAudio, focusing on **reliability** and **production readiness**.
- Engineered an **agentic decision-making engine** using **LangGraph** and Llama 3.1, enabling the assistant to maintain state across multi-turn conversations and handle **tool-calling** for local system tasks, demonstrating **function calling**.
- Optimized audio processing pipelines for near-real-time transcription and response, managing asynchronous streams between microphone input and LLM inference for **low-latency** user experience, a key aspect of **LLMOps**.
- Architected a **tool-integrated workflow** where the agent autonomously queries local files, checks system status, or performs web searches based on intent classification, showcasing **API orchestration**.
- Implemented custom **state-management logic for context retention**, allowing the assistant to handle interruptions and follow-up commands seamlessly, serving as **short-term memory**.
- Utilized a **vector database** for efficient retrieval of conversation history and user preferences, enhancing personalization and acting as **long-term memory**.
- Implemented comprehensive logging to track agent decision paths and **diagnose anomalies**, facilitating iterative **policy refinement** and establishing an **observability framework**.

EDUCATION

- **Galgotias University:** 2023 – Present
 - Bachelor of Technology in Computer Science and Engineering
 - Relevant Coursework: Artificial Intelligence, Machine Learning, Distributed Systems

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification
- Mathematics-Basics to Advance for Data Science and GenAI
- Machine Learning Crash Course: Fairness
- HCL GUVI Sql
- Introduction to Programming Using Python